Week, 26th Jan

Monday, 26 January 2026 10:32 AM

Lagrangian functions to define <u>microstates of system</u>

↳ radians & sec. (distance & time)

q → velocity $(\dot{q}_i)$ $\quad i = 1, \ldots, 3N$

L → units of energy

$L = KE - PE$

$$KE = \frac{1}{2} m_i \dot{q}_i^2$$

$$p_i = \frac{\partial L}{\partial \dot{q}_i} = m_i \dot{q}_i$$

↑ momentum

Hamiltonian is legendre transform of L (Lagrangian)

$H = KE + PE$ } total energy of the system

Phase space → q & p space

↑ momenta

→ Conservative system (energy conserved)

→ Isolated system (N, V,)

Macro —statistical mech.→ Micro

(6N dof)

↓

Experiments ↘

Data is here

Micro-canonical → divide system is smaller ensembles
& have initial conditions

↳ & instead of projecting / solving Lan. & equations for each particle]

use stats & subject collections from ensembles.

Q = # of members of the ensemble

f = phase space density (p, q)

→ Hamilton's equation of Motion

→ Liouville's Theorem

Getting rid of the dynamic evolution but sampling states at fixed times

↓

States w/ same energy will be sampled w/ the same frequency.

Postulate of equal a-priori probability

+ isolated, thermal equilibrium

+ time-independent ensemble average $\langle F \rangle$

→ Ergodic: time averages equivalent to just counting states
hypothesis (don't have to integrate any equations of motion)

All possible p, q that have the same E, are sampled w/ the same frequency

→ Global properties

Lagrange multiplier

P_ν → probability of microstates

Ω → # of microstates w/ given N, V, E

Fundamental eq.: $dU = TdS - pdV + \mu dV$

Boltzmann's Entropy formula ⇒ $\boxed{S = k \ln \Omega}$

→ Allowed microstates can have fluctuated volume

$\Delta S < 0$ → equally likely, and equal sample frequency

$\Omega(v) = v^N$

$\Omega(V) = V^N$

Enormous # of microstates even for a low entropy system

fluctuations → But the overall probability of this

fluctuations allow for this → But the probability of this happening is exceedingly small

(In the thermal limit)

> Entropy is a measure of accessible microstates